Indian Bank

Customer Behaviour Analytics

Identifying Behavioural Patterns Across Cards,
Transactions and Merchant Categories

NL_Banking.pdf · Graduation Project Submission

1,00,000	91.15%	5.38%	Rs.16,425	Rs.164.25 Cr
Total Transactions	Success Rate	Fraud Rate	Avg Transaction	Total Value

Dataset	4 CSV files — Customers (25,000) | Cards (32,457) | Transactions (1,00,000) | Merchants (500)
Period	January 2023 – June 2026 · 42 complete months
Primary Metric	Transaction Success Rate — 91.15% | Total Value Rs.164.25 Crore
Tools Used	Microsoft Excel (EDA & Charts) | Tableau Public (Interactive Dashboard)

1. Executive Summary

Headline findings · primary metric benchmarks · key gaps identified

Metric	Value	Benchmark	Assessment
Transaction Success Rate	91.15%	Industry: 85–92%	Within benchmark range
Overall Fraud Rate	5.38%	5,379 fraud cases	Action required
International Success Rate	84.65%	vs 91.50% Domestic	Critical gap: 6.85 pp
Average Transaction Amount	Rs.16,425	Median: Rs.3,148	Mean skewed by high-value
Total Transaction Value	Rs.164.25 Crore	42-month period	Jan 2023 – Jun 2026
Preventable Fraud	38.09% (2,049 cases)	Expired/Blocked/Lost card	Highest priority action
Platinum vs Classic Spend	Rs.20,546 vs Rs.14,298	43.7% premium	Card upgrade opportunity
RuPay Fraud Advantage	4.99% vs Visa 5.54%	55 bps difference	Network anomaly to investigate

Five Headline Findings

38.09% of all fraud is preventable. 2,049 fraud cases arise from expired card (18.73%), blocked card (12.97%), and lost card (6.34%) attempts. These are fully preventable through proactive card management alerts — no technology investment required.

International transactions have a 6.85 pp success gap vs domestic. 84.65% success rate vs 91.50% domestic, driven by an 11.36% decline rate on international transactions. This is the single most impactful performance gap in the entire dataset.

Merchant risk level is not a fraud predictor. Fraud rate is uniform at 5.36%–5.40% across Low, Medium and High risk merchants. Fraud is driven entirely by card status and customer behaviour, not merchant context. This finding directly contradicts the current fraud model assumptions.

Platinum cards average 43.7% more than Classic cards per transaction. Rs.20,546 vs Rs.14,298 — directly correlated with credit limits. Gold customers represent a large, identifiable upgrade target with a confirmed spend uplift on conversion.

Online Web + Debit Card is the highest failure combination at 9.77%. Nearly 1 in 10 transactions fails on this combination. Online Web also has the highest average transaction value at Rs.16,929, meaning the most valuable transactions face the highest failure risk.

2. Business Context, Problem Statement & Hypotheses

Dataset structure · primary metric decomposition · directional hypotheses

Business Context

The Indian banking ecosystem has become increasingly diverse, with customers interacting through multiple card products, payment channels, and merchant categories. The bank holds rich data across four interconnected datasets — Customers, Cards, Transactions, and Merchants — yet examining these independently fails to reveal the complete picture of customer behaviour and business performance.

Problem Statement

Which customer, card, transaction, and merchant characteristics are associated with meaningful differences in transaction success rates, spending levels, and fraud exposure? And what specific actions should the bank take based on these behavioural patterns?

Hypothesis Testing — Results Summary

#	Hypothesis	Dimension	Key Finding	Verdict
H1	Higher customer segments spend more per transaction	Customer Segment	Platinum spends 19.5% more than Standard (Rs.17,980 vs Rs.15,047)	✓ Confirmed
H2	36–45 age band drives the highest transaction volume	Age Band	36–45 leads with 25,078 transactions — highest of all age bands	✓ Confirmed
H3	Platinum cards carry higher average transaction amounts	Card Type	Platinum Rs.20,546 vs Classic Rs.14,298 — a confirmed 43.7% premium	✓ Confirmed
H4	Online Web channel shows higher failure rates	Transaction Channel	Online Web lowest success at 90.72% vs POS 91.24% — gap confirmed	✓ Confirmed
H5	International transactions fail at significantly higher rates	Is_International	84.65% vs 91.50% domestic — 6.85 pp gap driven by 11.36% decline rate	✓ Confirmed
H6	High-risk merchants correlate with elevated fraud rates	Merchant Risk Level	Fraud uniform: Low 5.38%, Medium 5.36%, High 5.40% — no correlation	✗ Rejected
H7	RuPay shows a distinct fraud profile vs other networks	Card Network	RuPay lowest fraud at 4.99% vs Visa 5.54% — 55 bps advantage confirmed	✓ Confirmed

3. Data Quality & Preparation

Source validation · issues identified · cleaning decisions

Dataset	Records	Issue Found	Action Taken
Customers	25,000	14 records with Annual Income = Rs.2 (data entry error)	Flagged as outliers; excluded from all income-based averages
Customers	25,000	2,539 records with 'Divorsed' spelling typo in Marital_Status	Treated as 'Divorced' in all analysis; raw value preserved in source sheet
Transactions	1,00,000	Fraud_Reason null for 94,621 non-fraud rows	No action — nulls are structurally valid and expected for non-fraud transactions
Transactions	1,00,000	Jul–Aug 2026 partial months (694 transactions)	Excluded from trend analysis; all charts and EDA cover Jan 2023–Jun 2026
Cards	32,457	6,204 customers hold 2–3 cards (multi-card scenario)	Valid business scenario; analysed separately in the Card Analysis section

Customer & Card Portfolio — At a Glance

Attribute	Profile	Attribute	Profile
Age Range	18–70 yrs | Mean: 41.8 | Median: 41	Gender	Female: 50.1% | Male: 49.9%
Customer Segment	Platinum: 33.4% | Gold: 32.3% | Standard: 24.8% | Basic: 9.6%	Account Type	Savings: 45.6% | Salary: 42.0% | Student: 8.2% | Current: 4.2%
Card Types	Gold: 30.7% | Silver: 27.9% | Classic: 22.6% | Platinum: 18.7%	Card Networks	Amex: 25.5% | RuPay: 25.1% | Visa: 25.0% | Mastercard: 24.5%
Card Status	Active: 93.9% | Expired: 3.1% | Blocked: 2.0% | Lost: 1.1%	Contactless	Enabled: 85.1% | Not enabled: 14.9%
Avg Customer Tenure	5.0 years | Range: 0 – 10 years	Cards per Customer	1 card: 75.2% | 2 cards: 19.8% | 3 cards: 5.0%

Data Quality Assessment: The dataset is clean and complete. All four tables join without orphan keys. The only substantive cleaning actions were excluding 14 erroneous income records and treating 2,539 spelling variants as 'Divorced'. All other nulls are structurally valid.

4. EDA — Customer Behaviour Analysis

Segment spending · age band patterns · gender · occupation

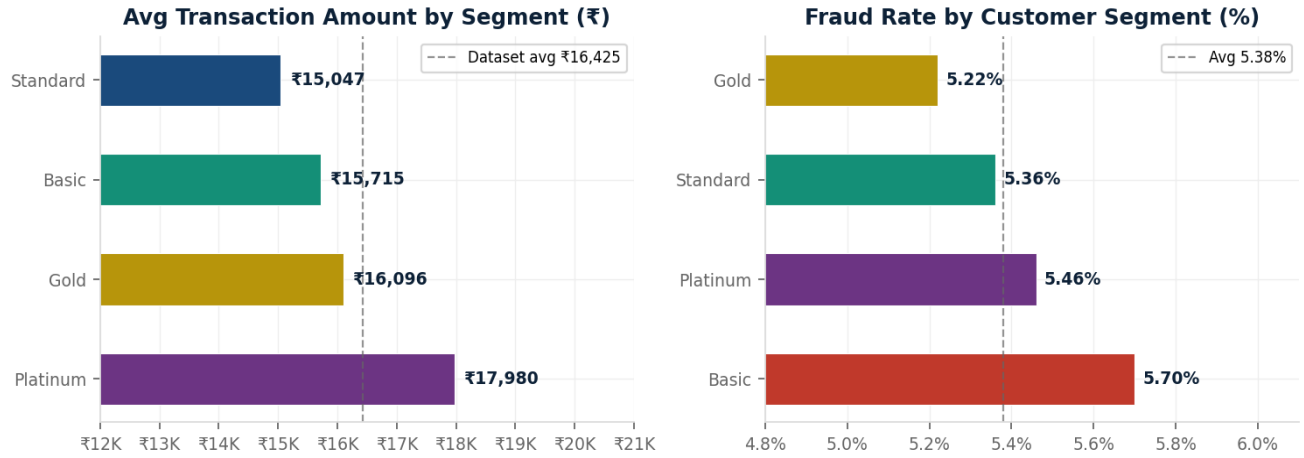


Figure 1 — Avg Transaction Amount by Customer Segment (left) | Fraud Rate by Customer Segment (right)

Key Insight: Platinum customers spend 19.5% more per transaction than Standard (Rs.17,980 vs Rs.15,047). However, fraud rates across segments are nearly identical (5.22%–5.70%), confirming that customer segment determines spending level but does not significantly predict fraud risk.

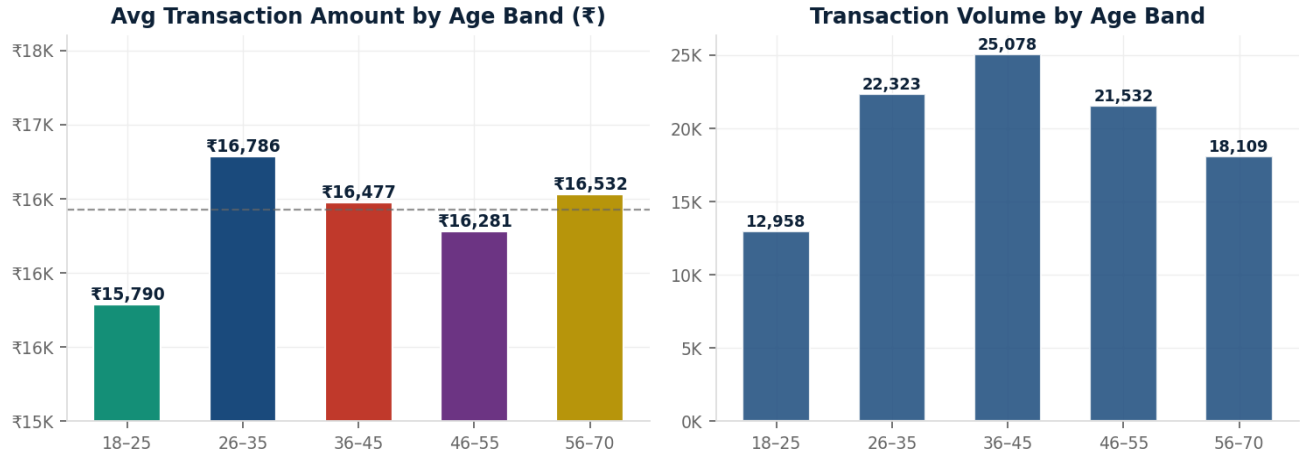


Figure 2 — Avg Transaction Amount by Age Band (left) | Transaction Volume by Age Band (right)

Key Insight: The 26–35 band spends the most per transaction (Rs.16,786) while 36–45 generates the highest volume (25,078 transactions). Female customers spend 5.9% more on average than male customers (Rs.16,895 vs Rs.15,953). Business owners and Chartered Accountants lead all occupations with Rs.18,424 and Rs.18,023 respectively.

5. EDA — Card Analysis

Card type spending · network fraud profiles · credit limit correlation

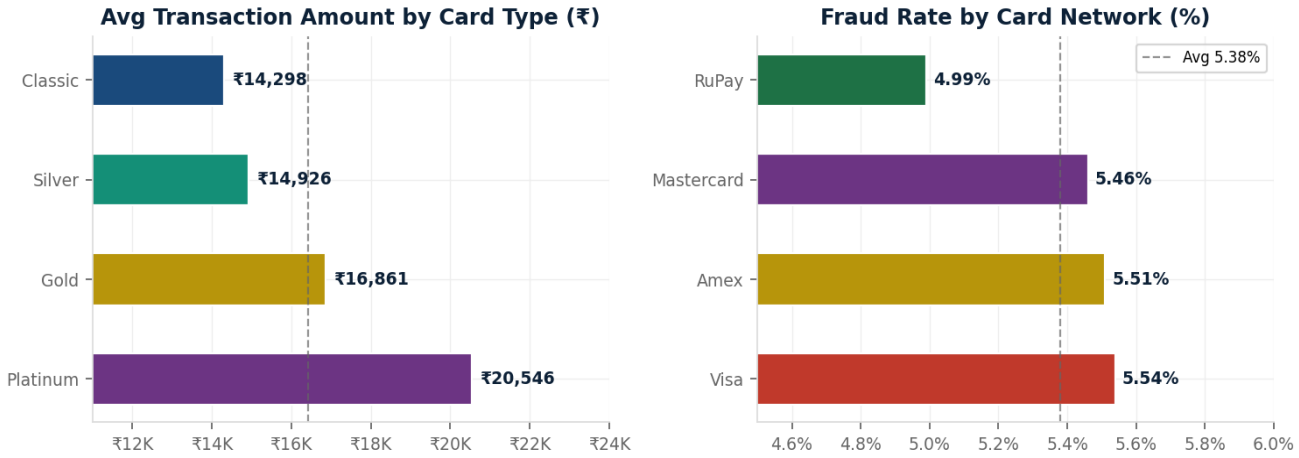


Figure 3 — Avg Spend by Card Type (left) | Fraud Rate by Card Network (right)

Key Insight: Platinum cards average Rs.20,546 per transaction — 43.7% above Classic (Rs.14,298). This premium directly correlates with credit limits: Platinum Rs.7L–Rs.15L vs Classic Rs.50K–Rs.1.5L. RuPay records the lowest fraud rate at 4.99% — 55 basis points below Visa (5.54%) — a meaningful and consistent gap across all channels.

Card Type	Avg Amount (Rs.)	Credit Limit Range	Fraud Rate	Gap vs Dataset Avg
Platinum	20,546	Rs. 7.0L – Rs. 15.0L	5.61%	+ Rs.4,121 (+25.1%)
Gold	16,861	Rs. 3.0L – Rs. 7.0L	5.30%	+ Rs.436 (+2.7%)
Silver	14,926	Rs. 1.5L – Rs. 3.0L	5.28%	– Rs.1,499 (–9.1%)
Classic	14,298	Rs. 50K – Rs. 1.5L	5.43%	– Rs.2,127 (–13.0%)

Card Network	Transactions	Avg Amount (Rs.)	Fraud Rate	Success Rate
Visa	24,910	16,627	5.54%	91.08%
American Express	25,536	16,494	5.51%	91.17%
Mastercard	24,378	16,310	5.46%	90.96%
RuPay	25,176	16,266	4.99%	91.40%

6. EDA — Transaction & Channel Analysis

Status distribution · channel performance · international impact · failure heatmap

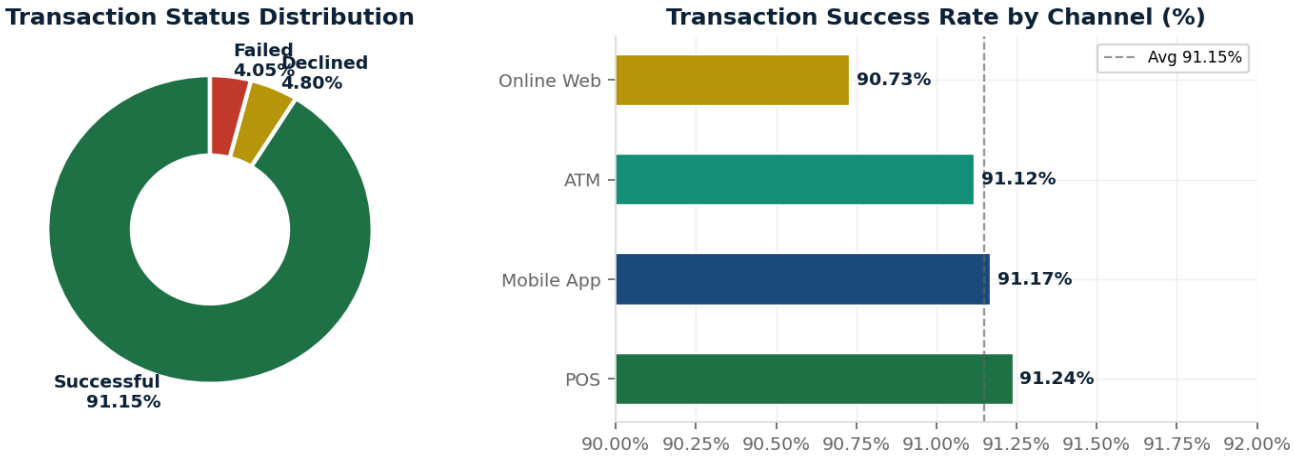


Figure 4 — Transaction Status Distribution (left) | Success Rate by Channel (right)

Key Insight: 91.15% of all transactions succeed. POS leads with 91.24% success and 48.3% of total volume. Online Web has the lowest success rate at 90.72% despite carrying the highest average transaction value at Rs.16,929 — indicating that high-value online purchases face the greatest authentication friction.

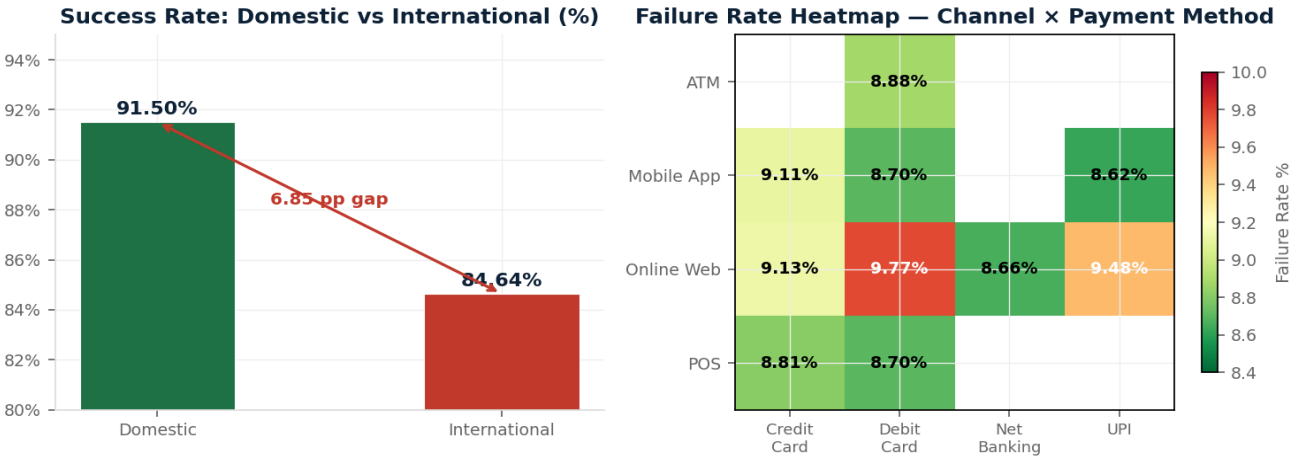


Figure 5 — Domestic vs International Success Rate (left) | Failure Rate Heatmap (right)

Key Insight: International transactions fail at a combined 15.35% rate (Declined 11.36% + Failed 3.99%) vs 8.50% domestic. The failure heatmap confirms Online Web + Debit Card as the highest risk combination at 9.77%. Blank cells in the heatmap represent genuinely non-existent combinations — ATM exclusively uses Debit Card; POS has no UPI or Net Banking usage.

7. EDA — Fraud Intelligence

Fraud drivers · preventable cases · channel and card type risk profiles

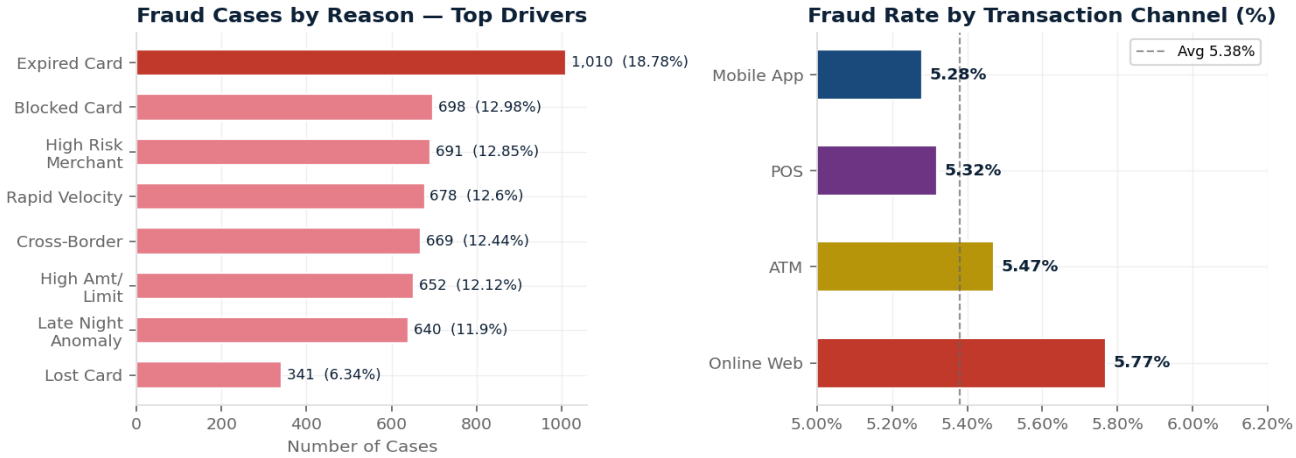


Figure 6 — Fraud Cases by Reason (left) | Fraud Rate by Transaction Channel (right)

Critical Finding: 38.09% of fraud is preventable — expired card (18.73%), blocked card (12.97%), and lost card (6.34%) attempts together account for 2,049 of 5,379 fraud cases. These arise from card management failures, not sophisticated fraud attacks. Proactive alerts alone would eliminate more than one-third of current fraud.

Dimension	Category	Fraud Rate	vs Avg (5.38%)	Cases
Channel	Online Web	5.77%	+0.39 pp	574
Channel	ATM	5.47%	+0.09 pp	872
Channel	POS	5.32%	–0.06 pp	2,571
Channel	Mobile App	5.28%	–0.10 pp	1,362
Card Type	Platinum	5.61%	+0.23 pp	1,041
Card Type	Classic	5.43%	+0.05 pp	1,234
Card Type	Gold	5.30%	–0.08 pp	1,636
Card Type	Silver	5.28%	–0.10 pp	1,468
Merchant Risk	High	5.40%	+0.02 pp	—
Merchant Risk	Low	5.38%	0.00 pp	—
Merchant Risk	Medium	5.36%	–0.02 pp	—

Counterintuitive Finding: Fraud rate is near-identical across all three merchant risk levels (5.36%–5.40%). Merchant risk classification is definitively not a reliable fraud predictor. Fraud detection resources should be redirected toward card status and customer velocity signals.

8. EDA — Spending & Merchant Analysis

Category value · transaction volume · total spend · strategic quadrants

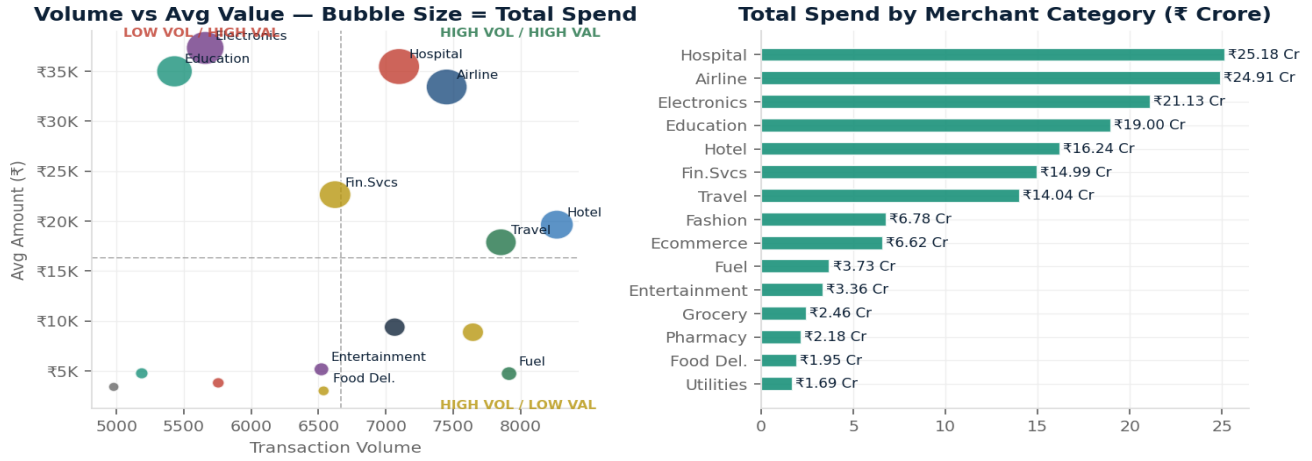


Figure 7 — Volume vs Avg Value Scatter Plot (left) | Total Spend by Merchant Category (right)

Key Insight: Electronics (Rs.37,333), Hospital (Rs.35,469), Education (Rs.34,990) and Airline (Rs.33,424) occupy the high-value quadrant — averaging more than double the dataset mean of Rs.16,425. Hospital leads total spend at Rs.25.18 Crore. Hotel, Fuel and Travel dominate transaction volume — the bank's three highest-frequency merchant touchpoints.

Merchant Category	Avg Amount (Rs.)	Total Spend (Rs.Cr)	Volume	Strategic Quadrant
Electronics	37,333	21.13	5,659	Low Volume / High Value
Hospital	35,469	25.18	7,098	Low Volume / High Value
Airline	33,424	24.91	7,452	Low Volume / High Value
Education	34,990	19.00	5,431	Low Volume / High Value
Hotel	19,632	16.24	8,270	High Volume / High Value
Travel	17,870	14.04	7,855	High Volume / Mid Value
Fuel	4,714	3.73	7,915	High Volume / Low Value
Food Delivery	2,986	1.95	6,538	High Volume / Low Value

9. EDA — Monthly Trends & Occupation Analysis

Transaction volume · fraud rate trends · occupation spending patterns

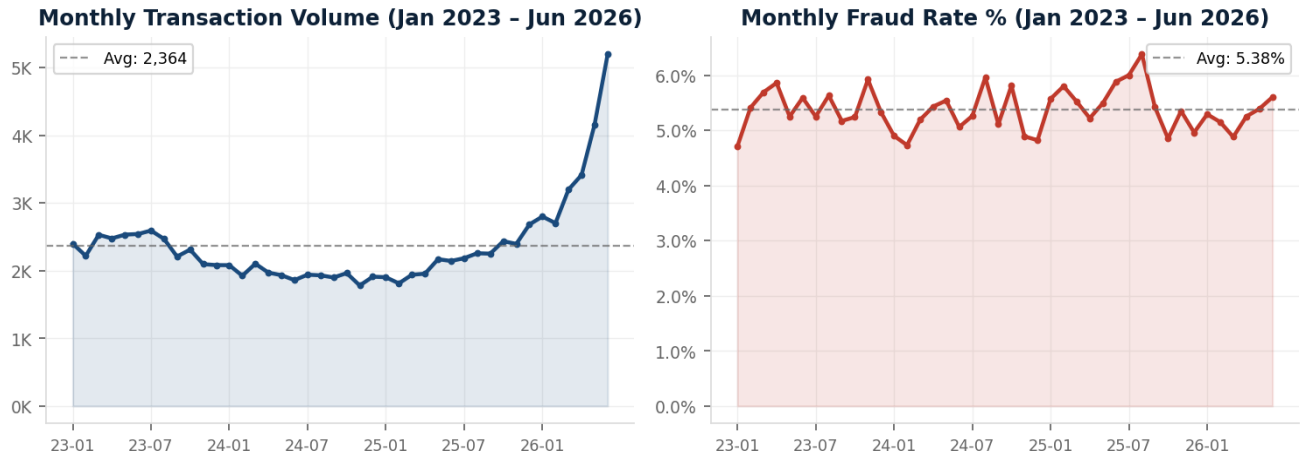


Figure 8 — Monthly Transaction Volume (left) | Monthly Fraud Rate % (right)

Trend Insight: Transaction volume shows consistent upward growth from 2023 through mid-2026. Fraud rate fluctuates within a narrow band of approximately 4.8%–6.0% with no sustained escalation — confirming that fraud is stable and not worsening over time.

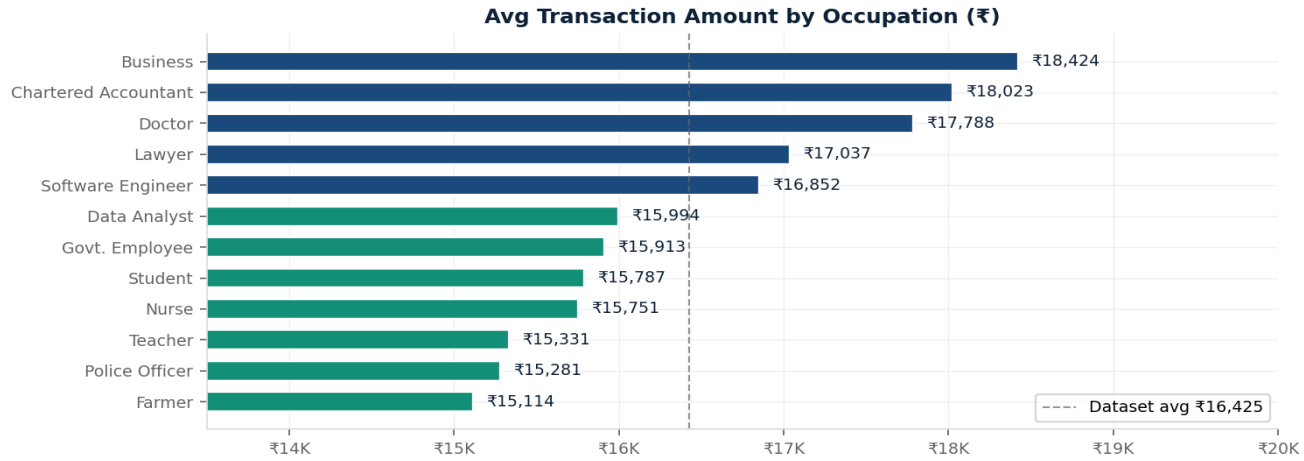
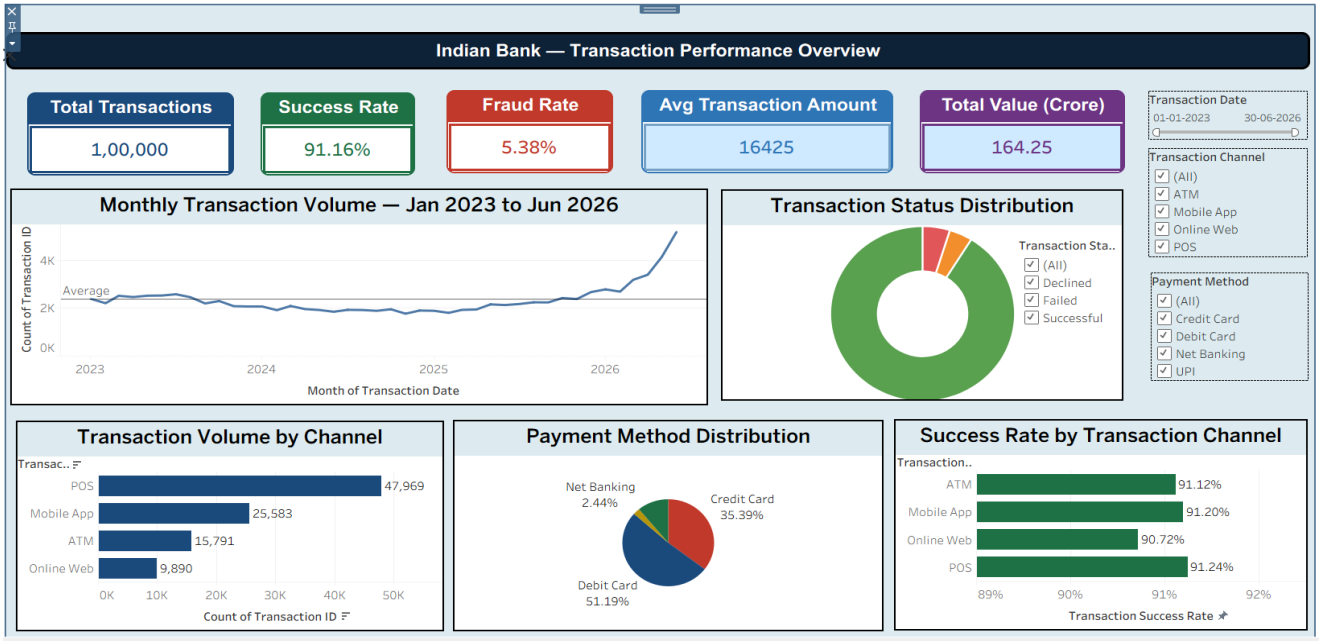


Figure 9 — Avg Transaction Amount by Occupation | Blue bars exceed dataset average of Rs.16,425

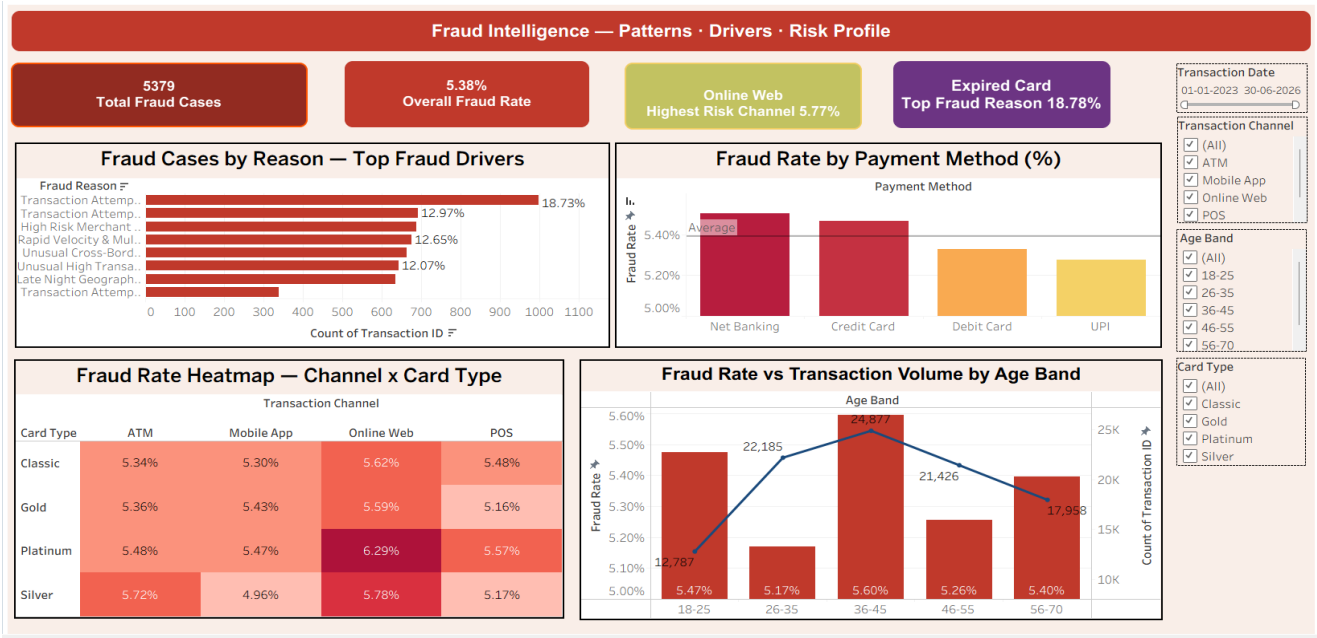
Key Insight: Business owners (Rs.18,424) and Chartered Accountants (Rs.18,023) record the highest average transaction amounts — approximately 22% above Farmers (Rs.15,114), the lowest occupation. The Rs.3,310 range across all 12 occupations is meaningful for targeted card product and merchant partnership strategies.

10. Interactive Tableau Dashboard

5 dashboards published on Tableau Public · Executive Overview & Fraud Intelligence shown below



Dashboard 1 — Indian Bank: Transaction Performance Overview | Filters: Date, Channel, Payment Method



Dashboard 4 — Fraud Intelligence: Patterns · Drivers · Risk Profile | Filters: Date, Channel, Card Type, Age Band

Tableau Dashboard: The complete interactive workbook includes 5 dashboards: Executive Overview · Customer Behaviour · Channel & Transaction Analysis · Fraud Intelligence · Merchant & Spending Analysis. All dashboards are published on Tableau Public with cross-dashboard filters and click-through actions for full stakeholder interactivity.

11. Key Findings

Data-verified insights · customer · card · transaction · merchant dimensions

F 1 Platinum customers spend 19.5% more (Rs.17,980) than Standard (Rs.15,047), with Platinum cards averaging Rs.20,546 — 43.7% above Classic. Credit limit is the primary spend driver.

→ *Gold customers (Rs.16,861 avg) are the prime Platinum upgrade target — data confirms a Rs.3,685 per transaction spend uplift on conversion.*

F 2 38.09% of all fraud is preventable — expired card (18.73%), blocked card (12.97%), and lost card (6.34%) attempts account for 2,049 of 5,379 fraud cases.

→ *Automated renewal reminders 60 days pre-expiry and real-time block push notifications would eliminate more than one-third of current fraud with no technology investment.*

F 3 International transactions succeed at only 84.65% vs 91.50% domestic — a 6.85 pp gap driven by an 11.36% decline rate. This is the single largest performance gap in the dataset.

→ *A pre-travel notification system and simplified 3DS for declared international travel would directly close the most impactful failure gap.*

F 4 Online Web + Debit Card fails at 9.77% — the highest failure combination. Online Web has the lowest channel success rate (90.72%) despite carrying the highest average value (Rs.16,929).

→ *Payment gateway optimisation — timeout handling, retry logic, and clearer authentication error messaging — would recover failures on the highest-value transaction channel.*

F 5 Fraud rate is uniform at 5.36%–5.40% across Low, Medium and High merchant risk levels. Merchant risk classification is definitively not a reliable fraud predictor.

→ *Fraud detection models must shift from merchant risk scoring to card status tracking, customer velocity monitoring, and cross-border anomaly detection.*

F 6 RuPay records the lowest fraud rate at 4.99% — 55 bps below Visa (5.54%) and 51 bps below American Express (5.50%). The gap is consistent across all transaction channels.

→ *Investigate whether RuPay's advantage stems from network-level controls or customer profile differences, and replicate the controls if structural.*

F 7 Electronics (Rs.37,333), Hospital (Rs.35,469) and Airline (Rs.33,424) average more than double the dataset mean of Rs.16,425. Hotel, Fuel and Travel lead in transaction volume (8,270 | 7,915 | 7,855) — together these six categories define the bank's premium and high-frequency merchant landscape.

→ *EMI and cashback on Platinum/Gold for the high-value trio; loyalty rewards for the volume trio. Two targeted programs covering the full merchant opportunity.*

12. Recommendations

Data-backed actions · prioritised by business impact · with implementation timeline

#	Area	Recommendation	Priority	Timeline
1	**International Authentication**	Deploy pre-travel alert system and streamlined 3DS for declared international travel — directly addresses 11.36% decline rate and the 6.85 pp success gap	**High**	0–30 days
2	**Card Renewal & Block Alerts**	Automated renewal reminders 60 days before expiry and real-time push notifications on card blocking — eliminates 38.09% of preventable fraud (2,049 cases)	**High**	0–30 days
3	**Online Web Gateway**	Audit checkout flow — implement retry logic, timeout fixes, and clearer authentication error messaging to address 9.77% failure rate	**High**	30–60 days
4	**Platinum Upgrade Program**	Target Gold cardholders (30,913 txns, Rs.16,861 avg) with Platinum upgrade offers — data confirms 21.9% spend uplift from Gold to Platinum adoption	**Medium**	30–60 days
5	**High-Value Category Offers**	EMI and cashback on Platinum/Gold for Electronics, Hospital and Airline — these categories average Rs.33K–Rs.37K vs dataset mean of Rs.16,425	**Medium**	60–90 days
6	**Fraud Model Reorientation**	Remove merchant risk level as a primary fraud signal (uniform 5.36%–5.40%) and refocus on card status, customer velocity and cross-border anomalies	**Medium**	60–90 days
7	**Loyalty for High-Frequency Categories**	Launch reward points for Hotel, Fuel and Travel — top 3 volume categories — to increase card stickiness and reduce payment method switching	**Medium**	60–90 days
8	**RuPay Fraud Investigation**	Investigate why RuPay shows 4.99% fraud vs 5.54% Visa — document root cause and replicate network controls if the advantage is structural	**Low**	90+ days
9	**UPI Adoption Campaign**	Promote UPI for Savings account holders — lowest fraud rate (5.26%) with growing share (11%) — improves experience and reduces fraud exposure	**Low**	90+ days

13. 90-Day Action Plan

Phased implementation · accountable owner · measurable success metrics

Phase & Priority	Action	Owner	Success Metric
Days 1–30 High Impact	Deploy international transaction pre-travel alert system with simplified 3DS flow	Digital Banking Team	International success rate improves from 84.65% toward 88.0%+
Days 1–30 High Impact	Implement automated card renewal reminders (60 days before expiry date)	Card Operations	Expired card fraud cases reduce by 50%+ (target: below 505 cases)
Days 1–30 High Impact	Activate real-time push notifications for all card blocking events	Card Operations	Blocked card fraud cases reduce by 50%+ (target: below 349 cases)
Days 30–60 Medium Impact	Audit Online Web checkout — implement retry logic, timeout handling, and error clarity	Technology Team	Online Web failure rate drops from 9.27% toward 8.50%
Days 30–60 Medium Impact	Launch Platinum upgrade campaign targeting Gold cardholders (30,913 eligible)	Product Team	5% of Gold customers upgrade; avg spend uplift of Rs.3,685 per transaction
Days 60–90 Medium Impact	Retrain fraud detection model — remove merchant risk; add card status & velocity signals	Risk & Analytics	Model precision improves; false positive rate reduces by 15%+
Days 60–90 Medium Impact	Launch EMI and cashback offers for Electronics, Hospital and Airline on Platinum/Gold	Merchant Partnerships	10% increase in transaction volume for these three high-value categories
Days 60–90 Medium Impact	Introduce loyalty reward programme for Hotel, Fuel and Travel transactions	Loyalty Team	Card usage frequency increases in top 3 volume categories by 8%+
Days 90+ Low Impact	Investigate RuPay fraud advantage; launch UPI adoption campaign for Savings account holders	Risk Research & Digital Payments	RuPay root cause documented; UPI share grows from 11% toward 15%

Implementation Note: The first 30 days focus exclusively on fraud prevention and international authentication recovery. These actions require no product changes — only operational communications — yet could reduce fraud by over 38% and meaningfully close the 6.85 pp international gap. The financial impact of these two actions alone justifies the entire analytical investment.

14. Conclusion

Summary of findings · strategic implications · path forward

Transaction Ecosystem Health

The bank's transaction ecosystem is fundamentally healthy. A 91.15% success rate across 1,00,000 transactions and Rs.164.25 Crore in total value over 42 months places the bank within industry benchmarks. Transaction volume has grown consistently from 2023 through mid-2026, and fraud rates show no sustained escalating trend.

The Three Most Actionable Findings

First, 38.09% of fraud is preventable through basic card management communications — renewal alerts and real-time block notifications. This is the highest-return intervention in the entire analysis: 2,049 fraud cases eliminated at near-zero cost. Second, international transactions fail at a 6.85 percentage point higher rate than domestic — a fixable authentication friction problem addressable with pre-travel alerts and simplified verification. Third, merchant risk level does not predict fraud. Realigning the detection model around this finding improves precision while reducing unnecessary merchant friction.

The Growth Opportunity

Platinum cards average 43.7% more per transaction than Classic cards. Gold customers represent a large, identifiable upgrade target with a confirmed spend uplift on conversion. Electronics, Hospital and Airline generate average transactions of Rs.33K–Rs.37K — more than double the dataset mean — making them the highest-priority categories for EMI and cashback programs. Hotel, Fuel and Travel lead in volume, making them the optimal channels for loyalty programs that increase usage frequency.

Path Forward

The 90-day action plan begins with fraud prevention and authentication recovery in the first 30 days, moves to channel optimisation and card upgrade programs in days 30–60, and concludes with model reorientation and high-value category partnerships in days 60–90. This phased approach delivers measurable results on the most critical issues while building toward the growth initiatives that will compound returns over subsequent quarters.

Strategic Takeaway: The most counterintuitive and most actionable finding: merchant risk level is not a fraud predictor. Fraud is driven by card status and customer behaviour. Realigning the detection model around this single finding would improve fraud prevention precision while reducing unnecessary merchant friction — a structural improvement achievable within 90 days.